Unit 4·6 Optimization (Enrichment)

Note Title 2014/04/20

Theme 5: Integration

Unit 5.1 Antiderivatives p 350

<u>Definition:</u> Function F is called an antiderivative of f on an interval if

$$F'(x) = f(x) \quad \forall \; x \in I$$

<u>Example:</u> ① $f(x) = x^2$

$\Rightarrow \quad F(x) = \frac{1}{3}x^3$

so $\quad F'(x) = x^2$

② $f(x) = \sin x$

$\Rightarrow \quad F(x) = -\cos x \quad$ etc.

But $\quad f(x) = x^2$ then

$F(x) = \frac{1}{3}x^3 + 4 \rightarrow$ and many more

So!

Theorem 1: p 351

F antiderivative, then the most general antiderivative of f on I is:

$$F(x) + C \rightarrow \text{arbitrary constant}$$

(1) $f(x) = \frac{1}{x}$ then $F(x) = \ln|x| + c$

(2) $f(x) = \sec^2 x$ then $F(x) = \tan x + c$

(3) $f(x) = \frac{1}{1+x^2}$ then $F(x) = \tan^{-1} x + c$

Table p 352.

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\sec^2 x$	$\tan x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec x \tan x$	$\sec x$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
$\frac{1}{x}$	$\ln\lvert x\rvert$	$\frac{1}{1+x^2}$	$\tan^{-1} x$
e^x	e^x	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$
$\sin x$	$-\cos x$		

1. If $g'(x) = 2x + 3x^{1.7} + \frac{1}{x}$ find $g(x)$.

$$g(x) = x^2 + \frac{3}{2.7} x^{2.7} + \ln|x| + c$$

2. $f'(x) = 3e^{x} + 7\underline{\sec^2 x} + \underline{\sec x \tan x}$

$f(x) = 3e^{x} + 7\tan x + \sec x + c$

3. $f'(x) = \dfrac{2+x^2}{1+x^2} = \dfrac{\cancel{1+x^2}^{\,1}}{\cancel{1+x^2}} + \dfrac{1}{1+x^2}$ *

$f(x) = x + \arctan x + c$

$\left(\tfrac{1}{2}, 1\right)$

4. $f'(x) = \dfrac{4}{\sqrt{1-x^2}}$ and $\underline{\underline{f\left(\tfrac{1}{2}\right) = 1}}$

$f(x) = 4\arcsin x + c$

$1 = 4\arcsin\left(\tfrac{1}{2}\right) + c$

$1 = 4 \times \frac{\pi}{6} + c$

$c = 1 - \frac{2\pi}{3}$

$f(x) = 4\arcsin x + 1 - \frac{2\pi}{3}$

5. If $f'(x) = (x^2+1)^7 x$ find $f(x)$.

$$f(x) = \frac{1}{16}(x^2+1)^8$$

$$f'(x) = \frac{1}{16} \cdot 8(x^2+1)^7 \cdot 2x$$

6. If $f'(x) = \dfrac{x^2}{\sqrt{x^3+1}}$ ✓ find $f(x)$.

$$= x^2(x^3+1)^{-1/2}$$

$$f(x) = \frac{2}{3} \times (x^3+1)^{1/2}$$

$$f'(x) = \frac{2}{3} \times \frac{1}{2}(x^3+1)^{-1/2} \cdot 3x^2$$

$$= \frac{3x^2}{2\sqrt{x^3+1}} \times \frac{2}{3} \checkmark$$

7. $$f'(x) = \frac{\sec^2\sqrt{1-x}}{\sqrt{1-x}}$$

$$f(x) = -2\tan\sqrt{1-x}$$

$$f'(x) = -2x - \sec^2\sqrt{1-x} \times \frac{1}{2}(1-x)^{-1/2} + c$$

$$= \frac{-\sec^2\sqrt{1-x}}{2\sqrt{1-x}} \times -2 + C$$

8. $$f'(x) = \frac{x^7}{\sqrt[3]{x^8+1}}$$

$$= \frac{1}{8}\left[8x^7\left(x^8+1\right)^{-\frac{1}{3}}\right]$$

$$f(x) = \frac{1}{8}\left[\frac{\left(x^8+1\right)^{2/3}}{\cancel{\frac{2}{3}}}\right] \times \frac{3}{2} + C$$

$$= \frac{3}{16}\left(x^8+1\right)^{2/3} + C$$